

Portfolio item 3: Report Skeleton

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1 Introduction

- Define transients in astronomy, focusing on SNe Ia, explaining why them being standardisable candles makes them important to study.
- Introduce the early excess flux SNe Ia and how studying correlations with these early excess SNe Ia light curves can lead to insights about the explosion environment.
- Mention how current standardisation methods leave a scatter in the residuals that appears to be systematic, thus can be corrected for, resulting in cleaner data for cosmological studies. Specifically, the limits of the Tripp equation and how it doesn't reduce the significance of multiple correlations with the supernova environments. By understanding the environments that produce specific light curves, not only can more be understood about the physics of the explosion, but this improved understanding can lead to improved standardisation methods.
- Introduce LSST, Rubin and Fink as the tools that will provide the means of data collection, the immense volume of alerts the survey will produce providing much better chances of finding early excess flux SNe Ia. Mention how the Fink broker will provide the processing means to manage the large data volumes and process the data to be readily used for analysis.

2 Supernova Ia physics and Progenitor Channels

2.1 Type Ia Supernovae

- Describe what type Ia SNe are, explain that they are thermonuclear explosions of carbon-oxygen (C-O) white dwarfs, and how they are standardisable candles, which leads into the standardisation section. Keep this section brief, refer to Ruiter and Seitenzahl (2024)

2.2 Standardisation

- Introduce the Tripp relation and how the SALT model is used to determine the colour and stretch parameter that characterise SNe through the empirical light curve fitting method known as SALT.

- Explaining the use of type Ia supernova as standardisable candles to make distance measurements that are useful in determining cosmological parameters such as the equation of state parameter for dark energy and via determinations of H_0, Ω_m, and Ω_Λ. Introduce the mass-step discovered by Sullivan et al. (2010) and then discuss the studies that have improved standardisation using correlations with more local properties such as lsSFR (Rigault et al., 2020),the connection to cosmological parameters (Brout et al., 2022) and how corrections for stretch and colour in SNe Ia need to be accounted for based on environment leading to a non linear correction (Ginolin et al., 2025).

2.3 Progenitor Channels

- Describe the two popular progenitor channels for type Ia supernovae, the single and double degenerate scenarios, and how they differ in the light curve data through modelling with different nickel 56 masses and distributions. Keep brief,

- Primary references are Deckers et al. (2022) and Ruiter and Seitenzahl (2024)

2.4 Early Flux Excess Supernovae

- Here will be an explanation of how important early observation, well before peak flux/magnitude of the supernova, is crucial in having light curves that can help determine the environment of the explosion and possibly the progenitor.

- Describe the early excess flux observed in a few of the SNe Ia and how they are characterised, such as through visual inspection and calculation of light curve residuals in mag and flux space.

- Discuss some of the proposed mechanisms that could explain the observed early excess in the light curves, such as ^{56}Ni masses and ^{56}Ni distributions, interactions with the SNe ejecta and the companion star, and interactions with the circumstellar material.
- Primary references will be Burke et al. (2022), Deckers et al. (2022), and Lim et al. (2023)

3 Host Galaxy Environment and SN Ia light curves

3.1 Host Galaxy Properties

- Discuss the host galaxy stellar mass step correction and proposed physical explanations for the observed correlation, as an extrinsic property like host galaxy mass is not going to directly affect the brightness of the SNe Ia.
- Describe the observed differences in correlations between global properties, such as specific star formation rate and more local properties such as the local specific star formation rate, and lower masses and higher SFR in the local environment correlates with SNe Ia properties, such as colour producing a more homogeneous population of SNe Ia .
- Primary references will include Ginolin et al. (2025), Kelsey et al. (2022), and Ramaiya et al. (2025)

3.2 The Environment and Early Excess Flux

- If early excess flux SNe Ia are found in specific environments, not only will that enable the study of the environments that early excess SNe Ia occur in. It will also assist in the cutting of SNe Ia in samples used in cosmology, where the early excess SNe Ia could contaminate the cosmology sample.
- The lack of a statistically significant sample size has made it so that early excess flux SNe Ia rates haven't been subject to population studies which makes it difficult to correlation the early excess SNe Ia with their environment properties.

- Primary references will include Burke et al. (2022) and Deckers et al. (2022)

4 Fink, Rubin and LSST

- Give an overview of the specifications of LSST and Rubin telescope, including details of the survey strategy used for LSST such as the few mini surveys and the area of the southern sky that will be covered by the WFD. Describe the volume of alerts that are expected per night as well. Leading to a discussion on why this volume of alerts requires a broker to process data for science.

- Describe the Fink broker and give an overview of how it helps deliver on the classification of SNe making enabling faster and simpler analysis of data. Describe how Fink's active learning approach to classifiers helps deliver on fast and accurate classification.

- Compare the ZTF survey data that Fink has been using to the data LSST is producing, and how through the higher cadence and greater depth of LSST will detect SNe Ia earlier thus creating statistically significant samples of EEx SNe Ia.

- Describe how Fink enables the selection of extra-galactic candidates using latitude cut offs and catalogue matching and host galaxy association again using catalogue matching. Then how SN are classified as candidates and how SN Ia candidates are found using the machine learning models trained on photometric data along with spectroscopic follow-up observation.

- Primary references for this section will be Fraga et al. (2024), Leoni et al. (2022), Möller and de Boissière (2020), and Möller et al. (2021)

5 Project Aims

- State the primary aim of the project, which is to use LSST data collected via the Rubin telescope to identify and classify early excess SNe Ia and then investigate how their incidence correlates with host galaxy properties.

- Creating a sample of transients that are extra-galactic using Fink
- From those transients, identify the SNe Ia in the Rubin data using the Fink broker through
    1. Spectroscopic classifications
    2. Machine learning classification algorithms such as SuperNNova
- Measure the SN Ia properties through SALT using the colour and stretch parameters.
- Identify excess flux candidates. The 18 per cent detection efficiency at low redshifts from Deckers et al. (2022) shows that we will need a large sample size to find the early excess SNe Ia.
- Connect excess flux candidates to host galaxies using DLR and find the correlation with the mass step.

6 Planned Methodology

- Use Fink, and its classifiers (if they are online) to start building a sample of SNe Ia.
- Follow the methodology of Deckers et al. (2022) to identify the early excess flux SNe Ia in the sample. Starting by using the SALT3 model to generated fitted curves. Then find exclusively positive flux residuals 6 days pre-maximum flux epoch calculated from the fitted curve.
- To assign SNe Ia with a host, we will use the Directional Light Radius method, which takes the separation from the SN to the nearest galaxy and then normalises it by the galaxy's directional light radius, which is the radius of a line drawn from a galaxy's centre to a nearby transient event such as a SNe Ia. Gupta et al., 2016
- Finally, compare early excess supernovae Ia rates to host galaxies' properties such as sSFR, galaxy stellar mass, and colour. We will also see how the colour of the SNe Ia varies across the sample to determine if any connections crop up with the host

properties, with the colour of early excess SNe Ia. This will be used to make conclusions about whether early excess SNe Ia happen in specific host galaxy environments.

7 References

- This will be were the references are!

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